Lesson plan · for the teacher

Two-step sign-in and passkeys

Cyber security · Level 2 · Lesson 3

Time
40 min

Materials
Board and chalk; Cards: one marked “password”, others marked with second steps; A paper “thief” sign (optional)

Learning goals

- Explain why a password alone isn’t enough.
- Name three kinds of proof, with an example of each.
- Explain why a code must never be shared.

Activity: Two locks: Stop the thief

1. Warm-up (5 min): Give one pupil the “password” card and the thief sign. Ask the class: if the thief knows the password, is the account lost?
2. Teach (8 min): Draw a key (something you know), a phone (something you have) and a fingerprint (something you are). Explain that two-step sign-in asks for two of them.
3. Stop the thief (15 min): Hold up each second-step card in turn. The class decides whether the thief can get past it. When you show “a friend asks for the code”, ask what the right answer is.
4. Share (7 min): Groups agree on the strongest lock and explain why. Write the class rule on the board: never give a code to anyone.
5. Wrap-up (5 min): Exit ticket: each pupil writes one kind of proof and why they never share a code.

Write “password” on one card. On other cards write: “code by text message”, “code from an app”, “fingerprint”, “backup code on paper”, “a friend asks for the code”. Make a simple “thief” sign from paper.

Discussion

- Why can’t a thief with your password get in?
- Which second lock is strongest, and why?
- What would you say to someone who asks for your code?

Exit ticket

- One kind of proof + why I never share a code.

Try it at school tomorrow
Play “Two locks”: give one pupil, the thief, a card marked “password”. The class suggests second locks (a code on the phone, a fingerprint, a backup code on paper) and decides which ones stop the thief, and why. Finish with the class rule: never give a code to anyone.

Practice questions · for you

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1 **A colleague can choose a text message code, an authenticator app or a passkey as a second step. Which is strongest?**

◯ A passkey ◯ An authenticator app ◯ A text message code

2 **Match each second step to what you should know about it.**

A code by text message	• • • •	Never approve one you didn't start
An authenticator app	• • • •	Works only on the real site
A sign-in prompt	• • • •	Can be redirected by a SIM swap
A passkey	• • • •	A new code every 30 seconds

3 **A real bank or mobile money company may phone you and ask for the code it just sent, to check it's you.**

◯ True ◯ False

4 **You're setting up two-step sign-in for the first time. Put the steps in order.**

☐ Turn on two-step sign-in with an authenticator app or a passkey

☐ Open your email account's security settings

☐ Do the same for WhatsApp, then mobile money

☐ Write down the backup codes and keep them at home

✎ **Which of your accounts doesn't have two-step sign-in yet? Which second step will you choose for it, and why?**

Check your answers

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1 **A passkey**

A passkey can't be phished or moved to another SIM: it works only on the real site.

2 **A code by text message — Can be redirected by a SIM swap**
An authenticator app — A new code every 30 seconds
A sign-in prompt — Never approve one you didn't start
A passkey — Works only on the real site

Knowing each step's weak point tells you how to use it safely, and why passkeys come out on top.

3 **False**

The code proves who you are to the system, not to a caller.

4 **Top to bottom: 2, 1, 4, 3**

Email first, because it can reset everything else.